

Risk Control as the Durable Edge: Loss Filtering, Profit-Armed Ratchets, and Market-Neutral Carry When Direction Fails

What is left to engineer once the directional signal is a coin flip

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Abstract

Two companion papers argued that short-horizon direction in liquid cryptocurrency perpetual futures cannot be predicted, and that most edges which appear to predict it are artifacts. If that is right, the interesting question is what remains to engineer. This paper answers with three risk-control mechanisms from the same multi-month program. None is a bet on the direction of the next move, and for each one the paper is explicit about the conditions under which it fails. The first is a volatility-extreme entry filter that declines to trade when a one-hour Bollinger reading is stretched. Of five candidate filters it is the only one that removes losers rather than winners, and a permutation test rejects the possibility that it merely dropped a lucky 139 trades, at a probability below one in ten thousand; on the program's directional book over June 2026 it lifts a plus 13.2 percent account return to plus 87.8 percent, though in the flat second half of the month even the filtered book bleeds. The second is a profit-armed ratchet, an account-level trailing stop that stays disarmed until equity is up a set amount and then locks a fraction of any further gain under a one-way high-water-mark. On a book that ran from 1,000 dollars to 3,461 and gave most of it back to 1,878, the ratchet locks about 2,926 dollars, and a full arm-by-trail grid shows the

lock is a broad plateau rather than a tuned point, failing only in a narrow corner of very tight trails and low arming. The third is a cross-sectional dollar-neutral funding-carry sleeve, the one market-neutral construction in the program that keeps its sign out of sample where cross-sectional momentum does not, at a net Sharpe near 1.0 over six years with beta close to zero. It does not pay in all conditions. A stationary block bootstrap puts its Sharpe at 1.01 with a wide ninety-five percent interval from 0.11 to 1.89; it loses in three of six calendar years; and a decomposition shows the steady funding coupon is only six to fifteen percent a year while the crowded-name price spread both makes and loses the rest. Over the exact June window where the ratchet was locking a windfall the carry sleeve was down 4.4 percent, so the two did not cover for each other. The contribution is a data-heavy account of what risk control buys once prediction is off the table. Two of the mechanisms defend a directional book, declining a share of its worst trades and holding onto a windfall that could not have been forecast; the third earns a modest market-neutral return that does not lean on the book at all. None of it amounts to a strategy that reliably makes money. Every number is reproduced by two deterministic scripts on the program's own data.

1. Introduction

Most of the effort in quantitative trading goes into the signal, the forecast of what happens next. When the forecast works, the rest of the system matters much less. The uncomfortable finding of the two companion papers is that, for short-horizon moves in these instruments, the forecast does not work: direction is not predictable out of sample (Dashyan, 2026a), and the handful of rules that looked predictive did not survive an adversarial re-examination (Dashyan, 2026b). That leaves a question most write-ups never get to, because a working forecast lets them step over it: if you cannot predict the move, is there anything left worth doing?

There is, and it is the part of the system that has nothing to do with prediction. This paper documents three mechanisms from the program that held up, and states plainly where each stops. A loss filter that refuses to enter when short-term volatility is stretched, so the book takes fewer of its worst trades. A profit-armed trailing stop that keeps a windfall instead of handing it back, without any view on whether the windfall will continue. And a cross-sectional funding-carry sleeve that earns a market-neutral return from the shape of the funding curve rather than from a directional call. None of the three forecasts the next move; all three are risk control or structure.

Read against the evidence that survives scrutiny, this is where whatever durable value the program has, modest as it is, seems to sit.

This version deepens an earlier and shorter treatment in three directions, each grounded in the program's own data rather than in argument. The loss filter is placed inside the full family of five candidate entry rules that were tested, so that the reader can see it is the exception rather than the norm, and its selection of losing trades is checked against a permutation null and against the monotone relationship between the entry extreme and the outcome. The ratchet is mapped across a wide grid of its two parameters and across four different account curves, which turns the claim of robustness from an assertion into a picture with a visible boundary. The carry sleeve is set against the cross-sectional momentum factor it competes with, decomposed into its funding and price components year by year, and subjected to the same block bootstrap and selection-bias accounting that the companion paper applied to the directional results. None of this added detail is meant to flatter the mechanisms. In several places it makes them look more limited than the shorter treatment did, which is what the extra scrutiny is for.

The framing matters because it inverts the usual hope. A reader coming from the companion papers might expect a rescue, a place where careful engineering recovers the edge that prediction could not deliver. That is not what happens here. Risk control cannot manufacture return out of a coin flip. The three mechanisms do something narrower than that. One of them declines a share of the losing entries. Another keeps a windfall the book stumbles into rather than letting it drain away. The last is not a defense at all but a market-neutral sleeve, indifferent to which way price goes, that earns a little on its own. Section 2 places the paper in the risk-management literature. Section 3 restates the program and its account model. Section 4 recaps why direction is the wrong lever, and adds the finding that the model's own confidence is no better a filter than its direction. Sections 5 through 7 take the three mechanisms in turn, reporting the full result for each and the conditions that bound it. Section 8 shows that neither the directional book nor the carry sleeve was all-weather, Section 9 combines them and states the ceiling, and Sections 10 and 11 discuss and conclude.

2. Background

The idea that risk control can be an edge in its own right is older than the machine-learning backtest. Kaminski and Lo (2014) asked when a stop-loss rule actually stops losses rather than

just cutting returns, and showed that the answer depends on the return process: a momentum-like series with negative tail dependence rewards stopping, while a mean-reverting one punishes it. The trailing stop studied here is a profit-armed version of that idea, aimed at a specific pathology, a run that front-loads its gains and then bleeds them back. That pathology is the one the momentum literature calls a crash. Daniel and Moskowitz (2016) documented how momentum strategies give back years of gains in short, violent reversals, and Barroso and Santa-Clara (2015) showed that scaling exposure by recent volatility, a purely risk-side intervention with no new forecast, recovers much of the strategy's lost Sharpe. Han, Zhou and Zhu (2016) put it more directly still, showing that a simple stop-loss overlay tames momentum's worst drawdowns. The common thread is that the downside of these strategies is more forecastable than their direction, so the room to add value is on the downside rather than in the call itself.

The behavioral literature explains why the loss filter and the ratchet are not free lunches that everyone already takes. Shefrin and Statman (1985) named the disposition effect, the tendency to sell winners too early and ride losers too long, which is the exact opposite of what a profit-armed ratchet and a loss filter enforce. A mechanical rule that holds winners under a ratchet and refuses losing setups is, in part, a device for overriding that bias. On the carry side, the sleeve here is a cross-sectional, dollar-neutral member of the carry family formalized by Koijen, Moskowitz, Pedersen and Vrugt (2018), applied to perpetual-futures funding rather than to the forward curves of currencies or commodities, and it competes directly with the cross-sectional momentum of Asness, Moskowitz and Pedersen (2013) and the time-series momentum of Moskowitz, Ooi and Pedersen (2012). Section 7 shows the carry sleeve keeping its sign out of sample where the momentum factor does not, which is the empirical reason to prefer it. That a cross-sectional factor works in crypto at all is unsurprising given the frictions and segmentation Makarov and Schoar (2020) document in these markets.

The empirical checks in this paper are borrowed wholesale from the data-snooping literature, because the companion paper insisted on them and it would be inconsistent to drop them here. Costs come first, since a factor is only real after them; Novy-Marx and Velikov (2016) show how many published anomalies shrink or vanish once realistic trading costs are charged, which is why the carry result is reported net of fees and tiered slippage and stress-tested to its breakeven cost. Then there is the search that produced a best-of-many result and inflates it, so the carry configuration is deflated for the number of settings tried using the reasoning of Bailey, Borwein,

Lopez de Prado and Zhu (2014) and the deflated Sharpe ratio of Bailey and Lopez de Prado (2014), and read against the reality-check tradition of White (2000) and Sullivan, Timmermann and White (1999). Serial dependence in daily returns inflates naive significance, so the carry Sharpe is given a confidence interval by the stationary bootstrap of Politis and Romano (1994) rather than by a textbook standard error, and Sharpe ratios throughout are read with the cautions of Lo (2002). The loss filter is checked against a permutation null in the spirit of the same tradition. None of these tools is novel. Turning them on one's own preferred result, rather than on someone else's, is the discipline the companion paper is about, and here it falls on results the author would rather have left clean.

3. The Setting

The program and its data are described in the companion papers; this section restates what the three mechanisms need and no more. The directional side trades twenty Binance USD-margined perpetual contracts. A language-model policy reads a large multi-source snapshot of each coin and issues a long or short call with a stated confidence, and outcomes are resolved on one-minute prices by an independent first-touch rule against fixed take-profit and stop-loss brackets, the same resolver used in Dashyan (2026b). Two bracket geometries appear. A symmetric bracket, take-profit and stop the same distance apart, breaks even at a fifty percent win rate. An asymmetric two-to-one bracket, take-profit twice as far as the stop, breaks even near a one-in-three win rate, and it is the geometry the program traded live. The illustrative directional book in Sections 5 and 6 is simulated at ten percent margin and eight times leverage, a shade more aggressive than the five percent and seven times of the companion papers, with a ninety-five percent capital cap, at most one open position per coin, a flat 0.1 percent round-trip fee, and compounding from a 1,000 dollar start over the whole of June 2026, from June 2 to June 29.

The directional evidence in this paper comes from two levels of that book, and the two need separating to avoid confusion later. The trade level is the set of resolved individual calls, 1,002 of them matured over the month, on which win rates and average returns are measured; this is the level at which the loss filter and its permutation test operate. The account level is the sequenced, capital-capped equity curve, on which position sizing, the ninety-five percent cap, and the one-per-coin rule all bind; this is the level at which the account returns and the ratchet operate. A filter that improves the trade-level average need not move the account curve by the same

proportion, because the cap and the sequencing reshape which trades are actually taken, and the paper is explicit about which level each number lives on.

The carry sleeve in Section 7 is a separate, market-neutral construction on the same twenty coins. At each rebalance it ranks the coins by funding rate, goes long the n lowest-funding names and short the n highest-funding names in equal weight, gross exposure held at 1.0 so the book is dollar-neutral and its beta to Bitcoin sits near zero. It is perps-only, with no spot leg. Its returns are computed net of a five basis point taker fee and a per-coin tiered slippage estimate, and the paper quotes both a full six-year backtest, 2021 through mid-2026 on fifteen-minute data, and the shorter recent window where the program actually ran. Every figure in the paper is reproduced by two deterministic scripts over this data, described in the Data and Code section, one for the carry stress test and the ratchet and one for the deeper permutation, bootstrap, and decomposition analyses added in this version.

4. Direction Is the Wrong Lever

The rest of the paper avoids the forecast, and the June book shows why. Almost all of the book's return is earned in the first four trading days, June 2 to June 5, when the market fell and a book that leaned short collected beta. After June 6 the tape stopped trending and the same book gave the gains back, trade after trade, at a win rate indistinguishable from a coin flip. The directional engine did not get better or worse in the second half of the month; it lost the beta that had been paying it, and with no real forecast underneath, there was nothing to replace it. Read that way, the June curve is a run of short beta that the calendar cut off, and the engineering question is not how to predict the turn but what to do once you accept that you cannot.

Before turning to what does work, one tempting shortcut has to be ruled out, that the model's own confidence could serve as the missing filter. It cannot. Sorting the month's fired trades by the language model's stated confidence gives a relationship that runs the wrong way. The trades the model was least sure of, confidence below sixty-five, won 55.8 percent of the time, while the trades it was most sure of, confidence eighty and above, won only 32.2 percent and lost 0.72 percent on average. High conviction was anti-predictive. The same holds for the model's own hard gates: of the eight wait conditions it applies, seven blocked trades that would on average have won, and only one blocked a losing set. The lesson for this paper is a narrow one. Neither the forecaster's conviction nor its own rules of thumb function as a risk control. The mechanisms

that follow work precisely because they ignore what the model believes and act on structure instead: where price sits in its band, how far the account stands below its peak, and the shape of the funding cross-section.

5. Mechanism One: The Entry-Filter Family

5.1 Five candidate filters, and only one removes losers

An entry filter never opens a position; it only refuses to open one. The program tested five, each blocking the trades that meet some adverse condition and keeping the rest, and the one filter that works can only be judged against the four that do not. Table 1 gives the counterfactual over the full month on the symmetric geometry, where the diagnostic is simple: a filter that blocks a set of trades with a below-baseline win rate is removing losers and helping, and a filter that blocks an above-baseline set is removing winners and hurting. The baseline book, every fired trade, won 62.8 percent of the time. Four of the five filters block sets that won more than that, from R3 at 64 percent up to R1 at 86 percent, so they cut winners and lower the return. Only R4, the Bollinger-extreme filter, blocks a set that won just 42 percent and lost 0.39 percent on average, and it is the only one of the five that raises the account (Figure 1).

Filter	Blocks	Blocked-set WR	Verdict	Kept ROI (5% size)
R1 BTC 2% reversal	21	86%	cuts winners	+27.9%
R2 D1 trend conflict	48	68%	cuts winners	+20.6%
R3 funding pay side	553	64%	cuts winners	+14.0%
R4 BB extreme	143	42%	cuts LOSERS	+60.1%
R5 adaptive high-vol	12	58%	cuts winners	+28.8%
Baseline (no filter)	0	62.8% (all)	reference	+27.7%
All five combined	660	62.6% (kept)	over-filters	+19.8%

Table 1. Five candidate entry filters over June 2026, symmetric geometry, trade level. A filter helps only if the set it blocks won less often than the 62.8 percent baseline. Four of five block winners; only R4 blocks losers.

Stacking all five removes 660 of about 1,100 trades and gives back most of the benefit.

The table carries two lessons beyond the identity of the winning filter. One is that filtering is not, in general, a good thing. The intuition that declining trades must reduce risk and therefore help is wrong here four times out of five, because most of the conditions that look prudent, a conflicting daily trend, an unfavorable funding side, a recent large move, happen to mark trades that were

about to win. Nor, it turns out, is more filtering better. Stacking all five removes 660 of the roughly 1,100 trades and leaves an account return below the single good filter, because the four bad filters drag down the one good one. The rest of this section is therefore about R4 specifically, and about establishing that its one useful property is real rather than a lucky draw.

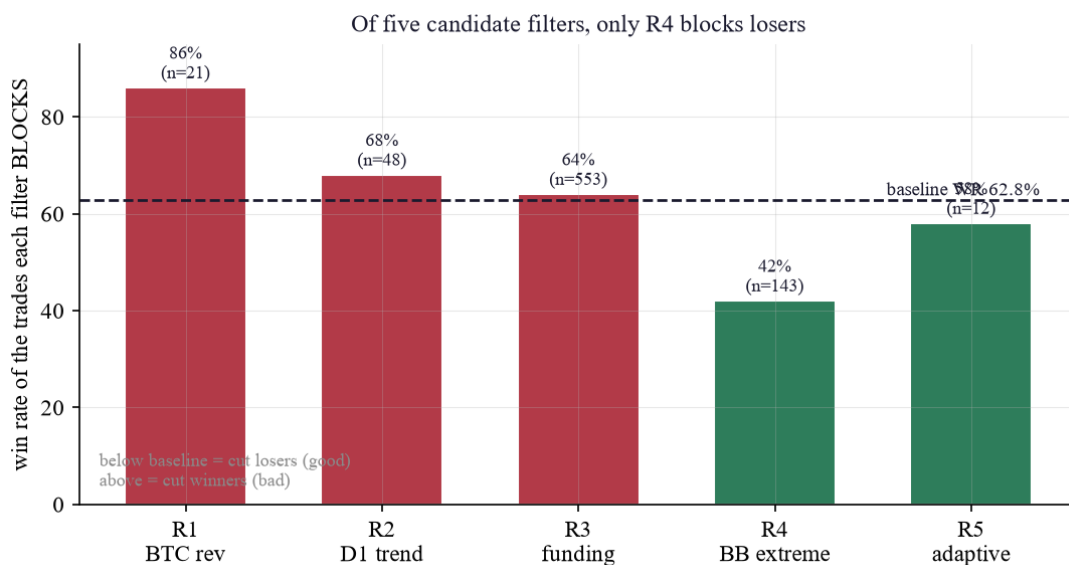


Figure 1. The win rate of the trades each of the five filters blocks, against the 62.8 percent baseline. A bar below the baseline line means the filter removed losing trades, which is what a useful loss filter should do. Only R4, the Bollinger-extreme rule, clears that test; the other four block sets that won more often than the book as a whole.

5.2 R4 on the account curve

R4 blocks a long when the one-hour Bollinger position sits above the upper band and a short when it sits below the lower band, on a standard twenty-period, two-standard-deviation band. It refuses the entries taken at a stretched extreme, where a directional call is most likely to be chasing a move about to snap back. Figure 2 shows the account effect on the native two-to-one geometry. The unfiltered book compounds a 1,000 dollar start to 1,132 dollars, a plus 13.2 percent return at a 59.8 percent win rate; the filtered book reaches 1,878 dollars, plus 87.8 percent at a 64.5 percent win rate, and its worst drawdown is smaller, 46.4 percent against 60.5 percent. On the flat symmetric bracket the same filter turns a minus 17.9 percent account into plus 35.0 percent. Table 1's kept return of 60.1 percent is not directly comparable to these figures, because it is measured at the companion-paper sizing of five percent margin on the

symmetric geometry rather than the ten percent, eight times, two-to-one setting used for the account curves here. The gain is purely defensive, and it comes from the trades the filter keeps the book out of.

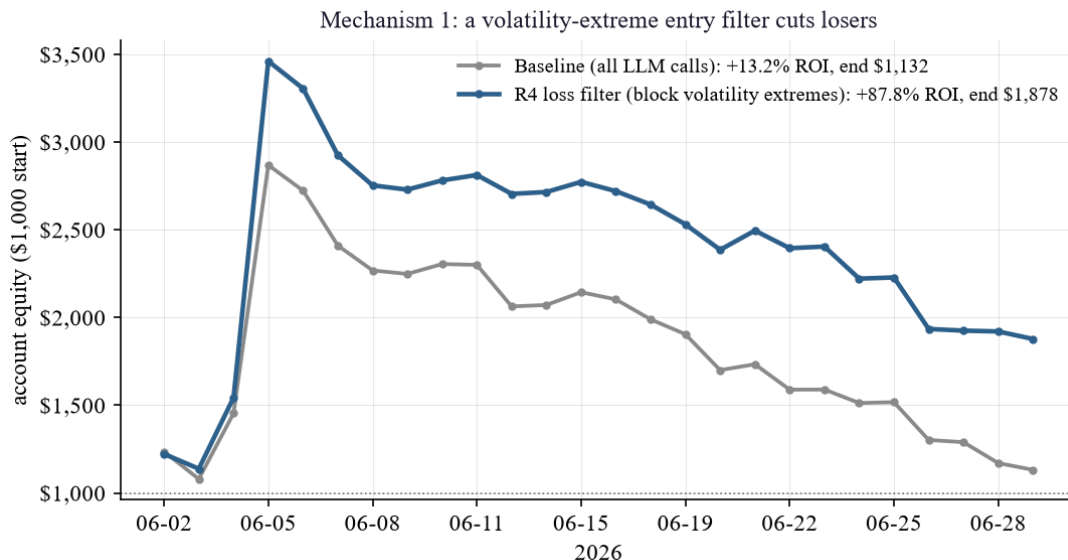


Figure 2. R4 on the directional book over June 2026, native two-to-one geometry, ten percent margin at eight times leverage. Declining the entries taken at a stretched one-hour Bollinger extreme lifts the account from plus 13.2 percent to plus 87.8 percent and cuts the worst drawdown from 60.5 to 46.4 percent. Because the filter only removes entries, the gain shown here is nothing but losses that were never taken.

5.3 Is it luck? A permutation test

The filter removes about 140 trades and the account improves, but so would dropping 140 trades that happened to include a run of losers by chance. The question is whether R4's specific selection is better than a random selection of the same size, and it has a clean answer. Of the 143 trades R4 would block across all fired calls, 139 matured within the month and enter this test. Take the 1,002 matured trades, and in each of twenty thousand draws remove a random 139 of them, recording the average return of the removed set. Under this null the removed trades are a representative sample, so their average return centers on the book average, plus 0.41 percent on the two-to-one geometry. R4's actual blocked set averages minus 0.25 percent, and on the symmetric geometry minus 0.47 percent against a null of plus 0.37 percent. Figure 3 shows the observed value sitting entirely outside the null distribution; in twenty thousand random draws none was as extreme, so the permutation probability is below one in ten thousand on both

geometries. The kept set improves by the mirror of the same amount, with the same probability. Whatever it is that R4 keys on, it is not just a fortunate draw of trades.

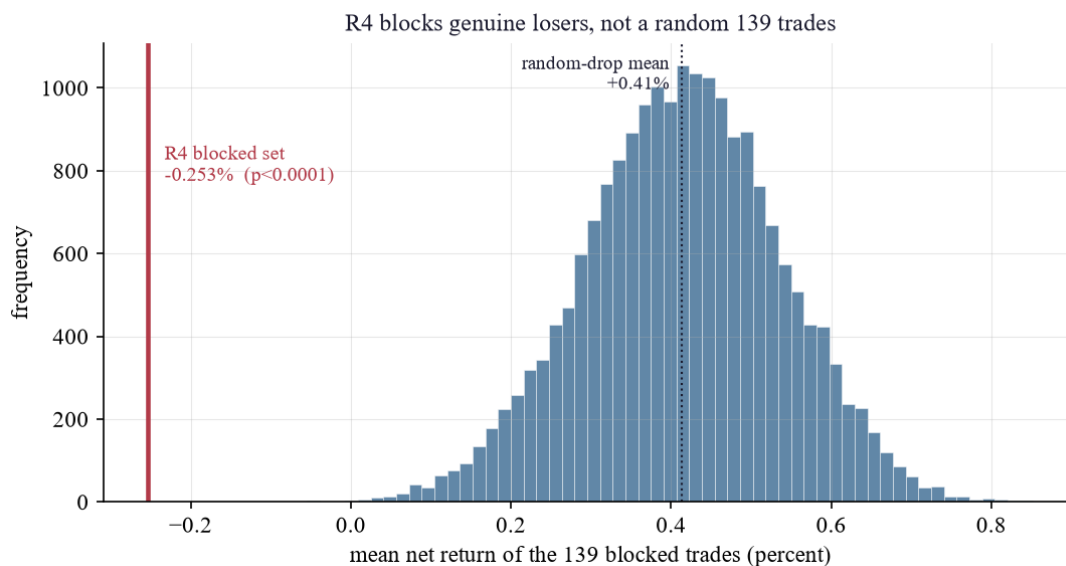


Figure 3. Permutation null for the loss filter, two-to-one geometry. The histogram is the average return of a random 139 blocked trades over twenty thousand draws, centered near the plus 0.41 percent book average. R4's actual blocked set, at minus 0.25 percent, lies entirely outside the null, for a permutation probability below one in ten thousand. The trades R4 blocks are losers in fact, not by chance.

5.4 Why it works: the entry extreme predicts the loss

The permutation test shows the selection is real but leaves open why it should be, and the reason is legible in the raw relationship between the entry condition and the outcome. Figure 4 sorts the month's short trades, the bulk of the book in a falling market, by their one-hour Bollinger position at entry and reports the win rate in each band. The trades entered with the band position below zero, meaning price was pinned at or below the lower band, are the losing bucket, winning only 48.3 percent of the time and losing 0.18 percent on average. As the entry position rises off the floor the win rate climbs, to 70, then 74, then 85 percent in the middle bands, before easing at the very top. Shorting into the lower band means selling into a level from which price tends to bounce, and that is exactly the trade R4 refuses. So R4 is not an accident fitted to June. It picks up a monotone and sensible relationship between where an entry sits in its band and how the trade resolves.

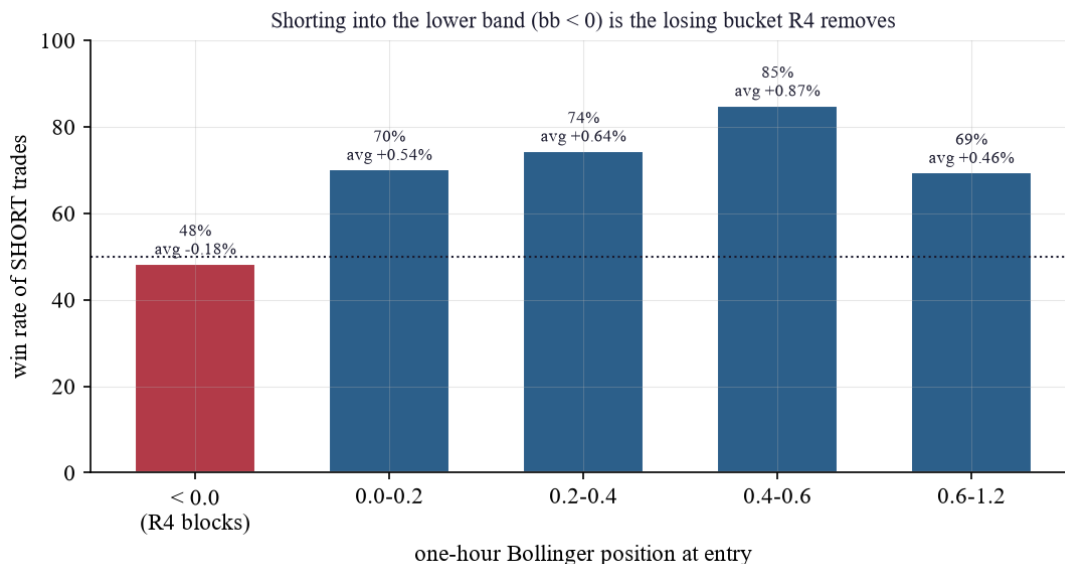


Figure 4. Win rate of the month's short trades by one-hour Bollinger position at entry. The lowest band, price below the lower Bollinger, is the losing bucket at 48.3 percent, and it is what R4 blocks for shorts. Win rate rises monotonically as the entry sits higher off the floor, which is what R4 keys on rather than a quirk of the month.

5.5 The limit: a loss filter cannot make a losing regime win

The improvement is real, but it has a ceiling, and that ceiling is on display in the last third of the month. Over June 20 to 29, in the chop that followed the early drop, the filtered book still lost money, minus 25.8 percent on the two-to-one geometry against minus 40.6 percent for the unfiltered book. The filter made a losing stretch less bad without making it a winning one. That is the correct expectation for a device that only ever subtracts trades: it raises the average quality of what remains, but if the remaining trades have no directional edge, filtering cannot manufacture one. In a bad regime the filter can only make the loss smaller; it cannot turn the regime into a winning one, and the account-level record over the full month, a large gain concentrated in the early trend followed by a bleed, is the direct consequence.

6. Mechanism Two: The Profit-Armed Ratchet

6.1 The pathology and why a naive stop fails

The second mechanism addresses the shape Section 4 described, a run that spikes and then bleeds. On the filtered June book the account climbs from 1,000 dollars to a peak of 3,461 on

June 5, then gives most of it back, ending the month at 1,878 for the plus 87.8 percent already quoted. About 1,600 dollars of realized peak equity evaporates after June 5. A naive account-level trailing stop cannot keep it, and the reason is instructive. Trail too tightly and an ordinary early wiggle stops you out before the run even happens. On this curve a six percent trailing stop is triggered by the routine 6.9 percent dip on June 3, locking the account at 1,138 dollars, a plus 13.8 percent that is worse than leaving the position alone. Triggered by noise, the stop is out of the position before the run it was meant to protect ever begins.

The fix is to arm the stop only after the account is convincingly in profit. The profit-armed ratchet stays disarmed until equity has risen by a set amount, an arm threshold, and only then begins trailing a fixed percentage under a one-way high-water-mark that never steps back down. Arming past the June 3 noise removes the false stop, so the run can happen, and only once the run is underway does the trailing stop become active to catch the reversal.

6.2 The lock, and its robustness

Figure 5 draws the three paths: the unmanaged bleed, the armed ratchet holding its lock, and the naive six percent trail dying on the early dip. The ratchet locks about 2,926 dollars, a plus 192.6 percent that more than doubles the plus 87.8 percent of the unmanaged book, and it recovers about two-thirds, 66 percent, of the roughly 1,600 dollars that would otherwise have bled away. The number that matters most is not the lock itself but its insensitivity to the two parameters. Figure 6 maps the locked value across a grid of seven arm thresholds and six trail widths. Across the great majority of that grid the lock is identical, 2,926 dollars, and it fails only in one corner, where a very tight trail of five or six percent is combined with a low arm threshold of twenty percent or less, in which case the account arms early enough and trails tightly enough to be caught by the June 3 dip and locks 1,138 dollars instead. Everywhere else, any arm of thirty percent or more with any trail, or any arm with a trail of eight percent or more, gives the same 2,926. That stability comes from how the mechanism is built rather than from a careful choice of settings. Arming after the run has started means the only stop that can fire is the one on the genuine reversal, and by then the high-water-mark is far enough above any sensible trail that the exact width no longer matters.

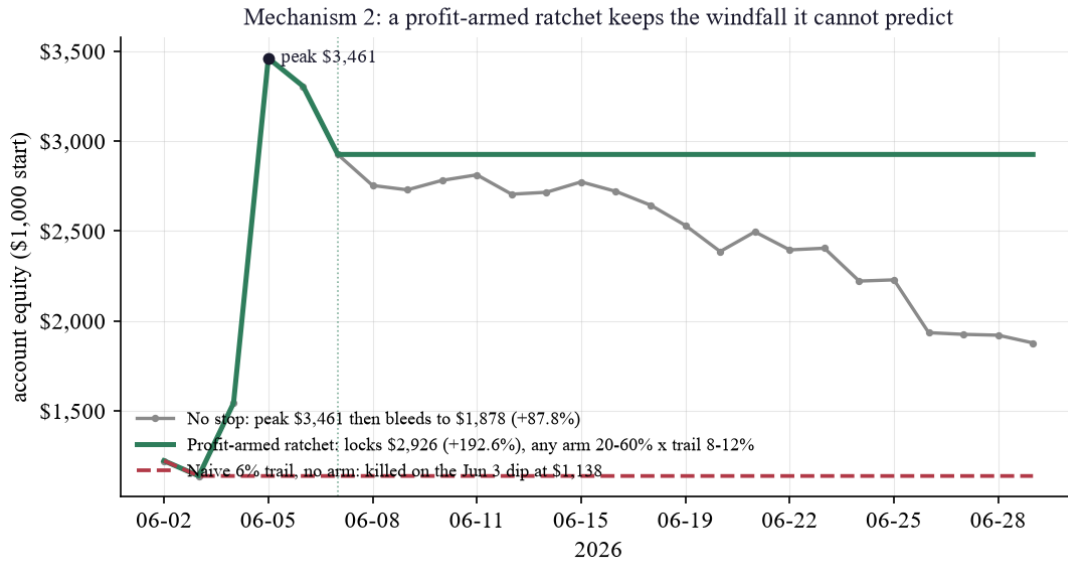


Figure 5. The profit-armed ratchet on the June book. Unmanaged, the account peaks at 3,461 dollars on June 5 and bleeds to 1,878. The ratchet locks about 2,926 dollars, and a naive six percent trail with no arming is stopped out on the June 3 dip at 1,138 dollars, worse than no stop at all, which is why the arming step matters.

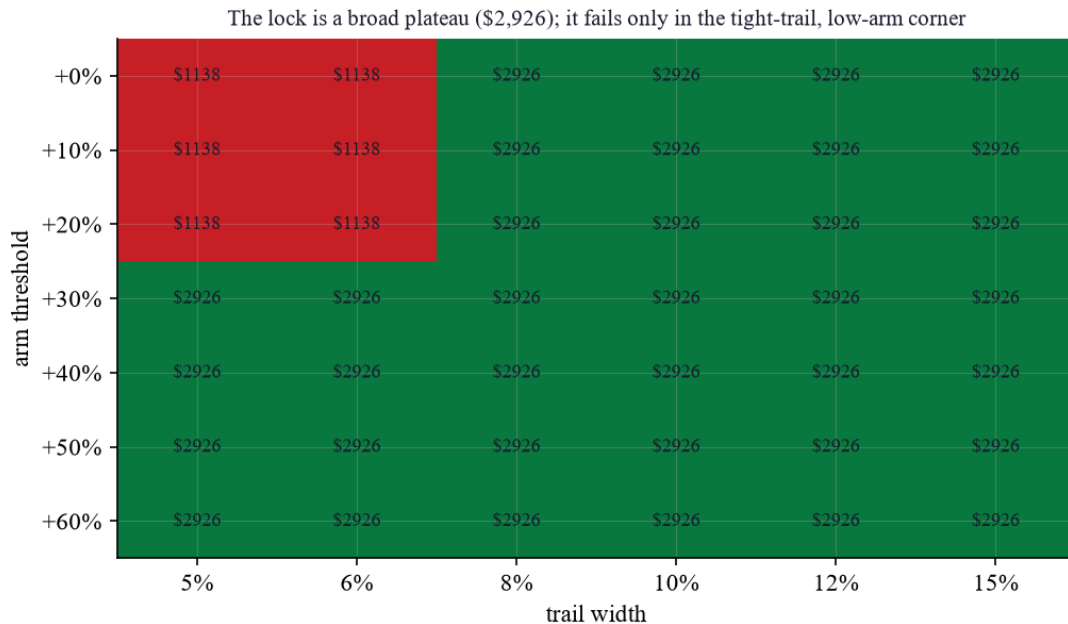


Figure 6. Locked account value across the ratchet's two parameters, arm threshold down the side and trail width across the top, on the June two-to-one curve. The lock is a broad plateau at 2,926 dollars. It fails only in the tight-trail, low-arm corner, where the stop arms early and trails tightly enough to be caught by the June 3 dip and locks 1,138. The flat plateau reflects the way the ratchet is constructed, not a fortunate setting of the two knobs.

6.3 The same rule on four different curves

A single curve can flatter any overlay, so the ratchet is worth applying to all four account curves the program produced, the filtered and unfiltered books on each of the two geometries, at one fixed setting of arm twenty percent and trail ten percent. Table 2 collects them, and the outcome is mixed, which is exactly what exposes how the mechanism really behaves. On three of the four curves the ratchet recovers between 66 and 78 percent of the give-back, turning even the losing unfiltered symmetric book, which ended the month at minus 17.9 percent, into a plus 95.7 percent lock. On the fourth, the unfiltered two-to-one book, it slightly underperforms doing nothing, locking 1,080 dollars against an unmanaged 1,132. The reason is precisely the arming lesson from the grid: that curve was already above the twenty percent arm threshold before its real run began, so the stop armed into the June 3 noise and was whipsawed out, the same failure the naive trail suffers. The ratchet is not universally beneficial. It rewards a book whose pre-run equity is below the arm threshold, and it can whipsaw a book that is already elevated when the arming triggers, which is a real constraint on how the arm level must be set relative to the curve.

Curve	Peak	Unmanaged end	Ratchet lock	Give-back recovered
Filtered, two-to-one	\$3,461	\$1,878 (+87.8%)	\$2,926 (+192.6%)	66%
Filtered, symmetric	\$2,755	\$1,350 (+35.0%)	\$2,364 (+136.4%)	72%
Unfiltered, symmetric	\$2,281	\$821 (-17.9%)	\$1,957 (+95.7%)	78%
Unfiltered, two-to-one	\$2,869	\$1,132 (+13.2%)	\$1,080 (+8.0%)	whipsawed

Table 2. The profit-armed ratchet at a fixed arm of plus 20 percent and trail of 10 percent, applied to all four account curves. It recovers most of the give-back on three, including rescuing the losing unfiltered symmetric book, but whipsaws on the unfiltered two-to-one curve, whose equity was already above the arm threshold before its run, so the stop armed into the June 3 noise.

6.4 A windfall kept, not made

The ratchet's usefulness runs out the moment the stop fires, and after that the account is flat. The ratchet keeps the windfall, but it cannot make another one, because after June 6 the directional edge is gone and there is nothing to re-enter into. Its recovered 66 percent is 66 percent of a give-back, not a new source of return, and it cannot catch the exact top in real time, which is why it locks 2,926 rather than the 3,461 peak. What the ratchet offers is worth having even so: given a

windfall you did not predict and cannot expect to repeat, it keeps most of it rather than letting it drain away. It is a device for the shape of the June curve, and that shape, a fast run of beta that bleeds out once the trend stops, is what a book with no forecast underneath will keep producing.

7. Mechanism Three: A Market-Neutral Carry Sleeve

7.1 The cross-sectional screen: why carry, not momentum

The first two mechanisms only ever defend a directional book. The third earns on its own, with no directional bet, and it is the one sleeve in the program whose net-of-cost return survives out of sample instead of dissolving into beta or fees. It did not win that place by default. The program screened the two standard cross-sectional, dollar-neutral factors on the same twenty coins, momentum and carry, and momentum looked the stronger of the two in sample. The best cross-sectional momentum configuration, long the thirty-day winners and short the losers, earned a full-period net Sharpe of 1.27 at an annualized 47.4 percent, with a beta of minus 0.05 and a t-statistic of 3.0. On the numbers a naive reader would deploy it. The out-of-sample split kills it. Its in-sample Sharpe of 1.60 collapses to 0.03 after the 2025 cutoff, and its worst drawdown over the period is 97 percent. Figure 7 puts the two side by side. Momentum is an in-sample artifact that does not survive the split; carry keeps its sign, holding a full-period Sharpe of 1.01 and an out-of-sample 0.90. Carry is the one to build on not because its headline number is larger, since it is not, but because it is the factor that is still standing after the out-of-sample split.

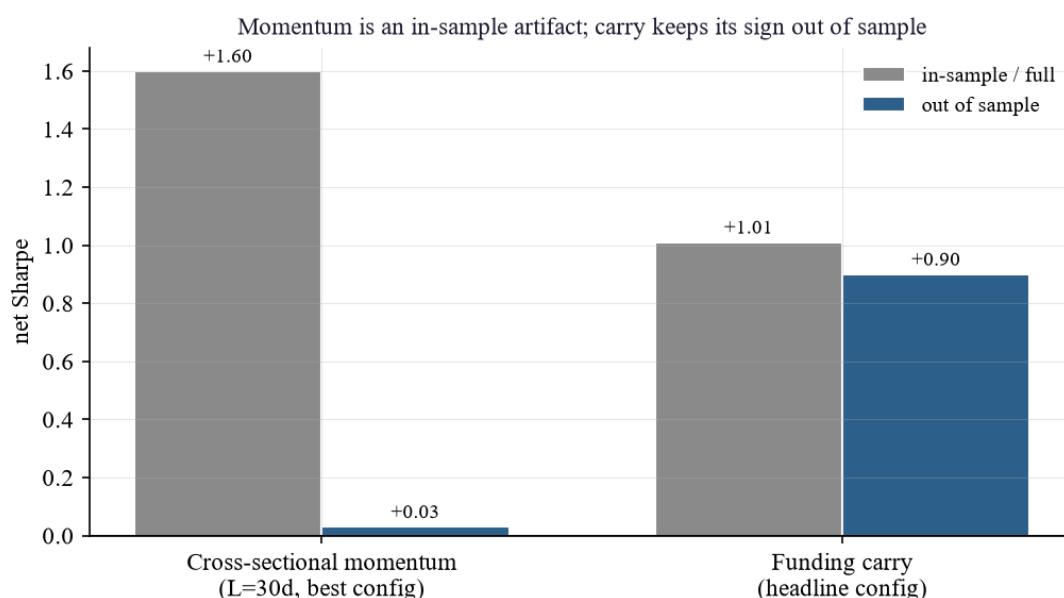


Figure 7. The two cross-sectional dollar-neutral factors, in sample against out of sample. Cross-sectional momentum posts a strong in-sample Sharpe of 1.60 that collapses to 0.03 out of sample, an in-sample artifact. The funding-carry sleeve keeps its sign, a full-period 1.01 and an out-of-sample 0.90. Out-of-sample survival, not the headline number, is why the paper builds on carry.

7.2 Construction and the six-year headline

The carry sleeve, as defined in Section 3, ranks the coins by funding rate and holds the lowest-funding names long against the highest-funding names short, dollar-neutral. It harvests the funding spread and, more than that, fades the crowded names that high funding marks, and that fade is where most of the return comes from. Reported over the full six years of data, 2021 through mid-2026, net of a five basis point fee and tiered per-coin slippage, the best configuration by net-of-cost Sharpe rebalances every three days on a one-day funding signal and earns a net Sharpe of 1.01, an annualized 39.1 percent, with an out-of-sample Sharpe of 0.90 and a beta to Bitcoin near zero.

7.3 The configuration sweep

The headline is one cell of a grid, and the grid is worth showing in full, both because it is the raw material for the deflation in Section 7.6 and because it makes the trade-offs visible. Table 3 gives all twelve configurations of the two lookbacks, two breadths, and three rebalance horizons. Two patterns run through it. Slower rebalancing lowers turnover and cost but also thins the sample and widens the drawdown, and wider breadth, five names a side rather than three, steadies the book at the cost of some return. The net Sharpes cluster between 0.44 and 1.01, the full-period best being the three-day, three-name cell in the headline. The out-of-sample column is noisier than the full-period column, as it should be on a shorter span, and several of the slower configurations show high out-of-sample Sharpes on very few rebalances, which is a small-sample effect rather than a reason to prefer them.

Lookback	Breadth n	Rebalance	Net Sharpe	OOS Sharpe	Max DD	Turnover
1d	3	1d	+0.86	+0.02	42%	0.96
1d	3	3d	+1.01	+0.90	46%	1.14
1d	3	7d	+0.48	+0.86	76%	1.22
1d	5	1d	+0.58	+0.12	50%	0.80
1d	5	3d	+0.88	+0.88	50%	0.97

1d	5	7d	+0.47	+1.91	64%	1.03
3d	3	1d	+0.96	+0.71	61%	0.53
3d	3	3d	+0.79	+1.05	66%	0.97
3d	3	7d	+0.44	+1.19	79%	1.09
3d	5	1d	+0.83	+1.01	52%	0.42
3d	5	3d	+0.76	+1.83	55%	0.81
3d	5	7d	+0.72	+2.05	62%	0.95

Table 3. All twelve carry configurations, full six years, net of five basis point fee and tiered slippage. The headline is the one-day-signal, three-name, three-day-rebalance cell, net Sharpe 1.01. The spread of Sharpes across the grid is the input to the deflation in Section 7.6; the high out-of-sample figures on the slow configurations rest on few rebalances.

7.4 Where the return comes from, year by year

The single most clarifying view of the sleeve is the decomposition of its return into the funding coupon and the price spread, year by year. Figure 8 splits the headline configuration's annualized contribution into the two parts. The funding coupon, the piece a perfectly price-hedged book would keep, is small and remarkably steady, between six and fifteen percent a year and positive in every one of the six years, including the losing ones. The price spread, the return from holding the legs of the crowded-name fade, is large and volatile: it added 136 percent in 2021 and 37 to 39 percent in the two good recent years, and it subtracted 19, 23, and 29 percent in 2022, 2023, and 2026. That contrast is the whole story of the sleeve. Its reliable component, the funding coupon, is small, while the component that drives the returns, the price spread, is the one that cannot be counted on. Figure 9 shows the same story at the level of the total, a genuinely positive six-year record that is negative in three of the six calendar years.

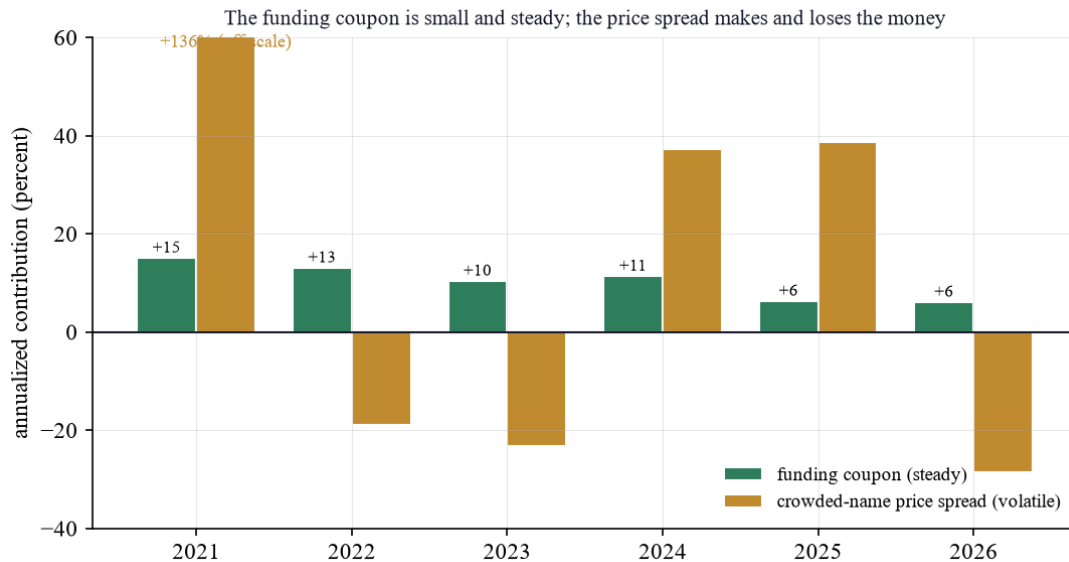


Figure 8. The carry sleeve's annualized return split into the funding coupon and the crowded-name price spread, by year. The coupon is small, six to fifteen percent, and positive every year. The price spread is large and swings from plus 136 percent in 2021 to minus 29 percent in 2026. The steady part is small; the large part is what makes and loses the money.

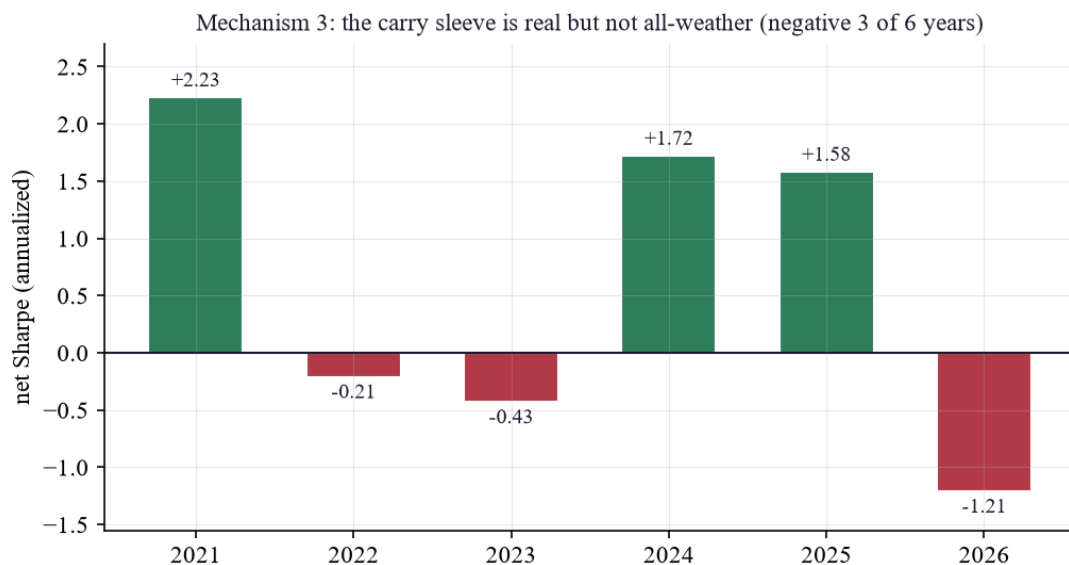


Figure 9. Net Sharpe of the carry sleeve by calendar year, headline six-year configuration, net of fee and tiered slippage. Positive over the whole sample but negative in three of six years and in a drawdown in 2026. It is a sleeve for a dispersion regime, not one to run in all markets.

7.5 Cost and capacity

Because the sleeve rebalances often, transaction cost decides whether it is real, and it deserves pressing. Figure 10 sweeps the assumed slippage from zero up to thirty basis points per side on top of the fee. The annualized return stays positive across the entire sweep, and the out-of-sample Sharpe stays positive out to about fifteen basis points, where it has fallen to roughly 0.3, and crosses zero near twenty. The realistic tiered estimate for this book is on the order of five to six basis points per side, comfortably inside the survivable range, and a maker fee of two basis points instead of the taker five lifts the net Sharpe from 1.01 to 1.12. The carry sleeve survives the costs a real desk would pay on the sizes tested, which is more than most of the directional machinery in the program can claim. It does not follow that it survives at arbitrary size; capacity, the point at which the crowded-name spread is arbitrated away by others running the same screen, is not modeled here and is a separate limit on deployment.

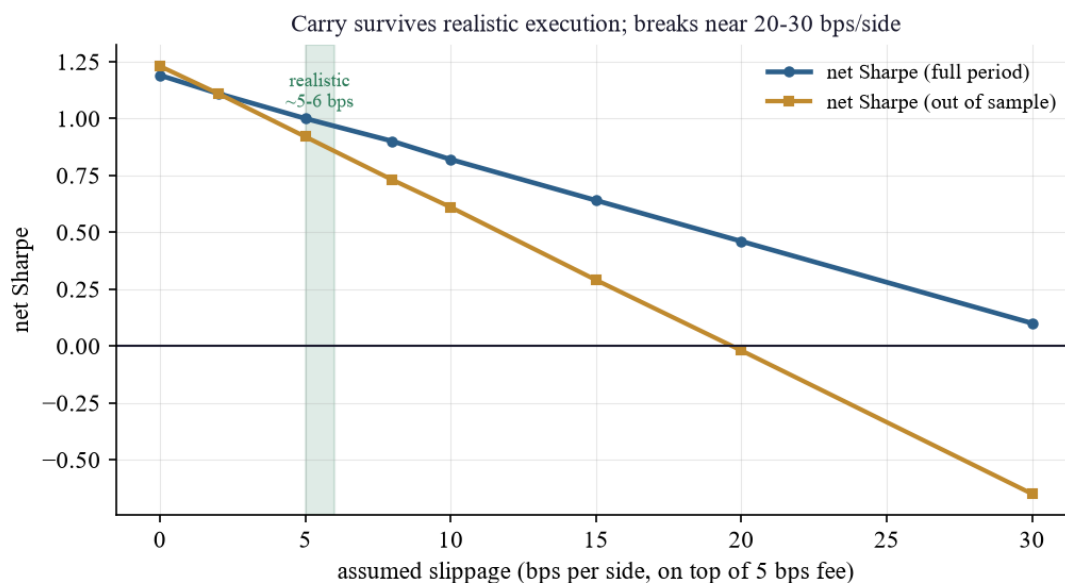


Figure 10. Breakeven slippage for the carry sleeve, headline six-year configuration. The annualized return stays positive across the whole zero-to-thirty basis point sweep, and the out-of-sample Sharpe holds above zero out to about fifteen basis points per side. The realistic tiered cost for this book, roughly five to six basis points per side, sits well inside the survivable range.

7.6 How real is the Sharpe? Deflation and a bootstrap interval

Two adversarial checks stand between the headline Sharpe and belief, both borrowed from the companion paper. The first is that 1.01 is the best of twelve configurations, so some of it is the

search rather than the signal. If the twelve are treated as independent trials with a common true Sharpe, which is conservative because they are in fact highly correlated, the expected best-of-twelve under pure selection is about 1.06, slightly above the observed 1.01. That comparison is deliberately harsh: the true number of independent trials is far below twelve, so the real selection bar is lower and the observed Sharpe likely clears a properly deflated one, but the exercise makes the point that the full-period 1.01 is not the number to lean on. The out-of-sample 0.90, measured on data no configuration was chosen on, is the clean read, and it is close enough to the in-sample figure to suggest the selection inflation is modest. The second check is serial dependence. A stationary block bootstrap of the headline configuration's daily returns, resampling blocks of about four days to preserve autocorrelation, puts the Sharpe at 1.01 with a ninety-five percent interval running from 0.11 to 1.89, and a probability of a non-positive Sharpe of about 1.2 percent. The edge clears conventional significance levels, but the interval is wide, and the whole of it is reported here because the point estimate alone would overstate how precisely the Sharpe is known. The carry effect appears to be real even if its magnitude is poorly pinned down.

7.7 The favorable-window mirage

A shorter, more recent window flatters the sleeve considerably, and it is worth working through why the prettier number misleads. On 2024 to 2026 data with a wider book, five names a side rebalanced weekly, the same construction shows a net Sharpe of 1.62 and a worst drawdown of only 11 percent, far prettier than the 46 percent drawdown of the six-year headline. That better number is a favorable-window slice, and reporting it as the result would be precisely the selection bias the companion paper warns against, the same trap that Section 7.1 caught cross-sectional momentum in. Both numbers are real; the six-year one is the one to believe. The recent window looks better because 2024 and 2025 were good carry years and the slice happens to exclude the losing ones, which is exactly the kind of window a hopeful researcher chooses without noticing.

8. Neither Mechanism Is All-Weather

It would be tidy if the directional book and the carry sleeve were complementary, the sleeve earning in the flat months where direction is a coin flip and the directional beta paying in the trending ones. The program ran both, so the claim is checkable rather than assumed, and the

check does not support the tidy story. Over the exact June window where the ratchet was locking a windfall on the falling tape, the carry sleeve was losing. Figure 11 overlays the two, each normalized to the start of June. The directional book spikes on the June 2 to 5 drop and then bleeds; the carry sleeve drifts steadily down, ending the overlap window at minus 4.4 percent. The month that was best for the directional beta was a mildly losing month for carry.

This overlap in their bad months is not an accident that better engineering could remove; it follows from how each mechanism earns. The directional book's good days are trending days, and a sharp directional move is often exactly the kind of low-dispersion, one-way regime in which the cross-sectional carry spread compresses. The two are not reliable hedges for each other. A portfolio that runs both will sometimes get the sum of their good regimes and sometimes, as in June, the directional windfall alongside a carry drawdown. Stating that outright is more useful than reporting a correlation number, which would suggest the diversification is dependable when a single shared month already shows it is not.

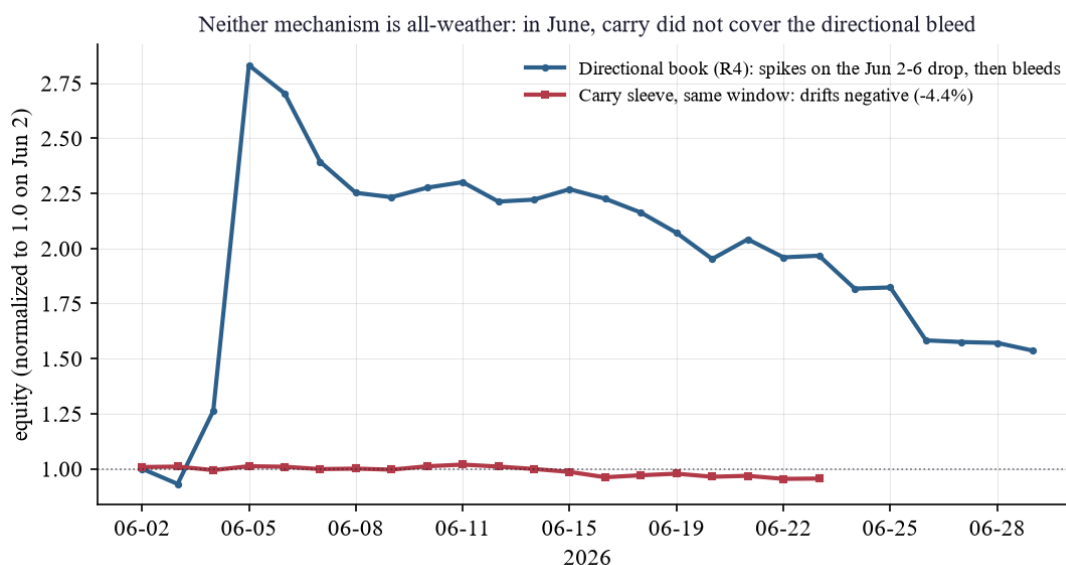


Figure 11. The directional book and the carry sleeve over the same June window, each normalized to 1.0 on June 2. The directional book spikes on the early drop and bleeds; the carry sleeve is down 4.4 percent over the window. The best month for directional beta was a losing month for carry, so the two did not cover for each other.

9. The Combined System and Its Ceiling

Put the three mechanisms together and the shape of the achievable system is clear, and so is how far it can go. Table 4 lays them side by side. Each does one narrow thing well, and the limits of one are not patched by the other two.

Mechanism	What it does	Headline	Where it breaks down
Loss filter (Section 5)	Declines entries at stretched volatility extremes	+13.2% to +87.8% on the June book; blocked set won 42%, permutation p below $1e-4$	Only subtracts trades; the filtered book still loses in chop
Profit-armed ratchet (Section 6)	Locks a windfall under a one-way high-water-mark	Locks about \$2,926 (+192.6%), a broad plateau across arm and trail	Goes flat after the lock; whipsaws if armed above the pre-run equity
Carry sleeve (Section 7)	Dollar-neutral funding carry, beta near zero	Net Sharpe 1.01 over six years, OOS 0.90, bootstrap CI 0.11 to 1.89	Negative in 3 of 6 years; steady coupon only 6 to 15% a year; not a hedge for the book

Table 4. The three risk-control mechanisms, what each contributes, and where each stops. None forecasts the next move; the value of all three together is bounded by the fact that none of them creates a directional edge.

The ceiling is the real conclusion of the paper, and overselling the combination would betray it. Stacked together, the mechanisms give a system that turns away the worst directional entries and holds onto most of whatever windfall the market's beta happens to deliver, with a market-neutral sleeve alongside them earning a net annualized return near forty percent over the full six years. That figure includes three losing years, and the sleeve loses outright in low-dispersion markets. That is a real, deployable structure, and it is emphatically not a money machine. Its directional component still depends on beta it cannot predict, and its neutral component still has losing years and a wide confidence interval. Risk control reshapes the distribution of outcomes without adding any predictive expectation to it. The book takes fewer of its worst trades, gives back less of a windfall once it has one, and carries a small uncorrelated return stream beside the directional book. For a program that started by trying to predict direction and found it could not, that is a more useful place to end than another overfit forecast.

10. Discussion and Limitations

The examples come from one program on twenty instruments over particular windows, so the specific numbers are illustrative rather than universal. The directional and ratchet results rest on a single month, June 2026, which is a small sample for an account curve and a trending-then-

chopping regime that will not always be the shape on offer. The permutation test that validates the loss filter is a strong result for that month, twenty thousand draws with no random selection as extreme, but it speaks to the resolved trades of a single month and says nothing about the next one. The ratchet's robustness across its settings is a property of any front-loaded spike-then-bleed curve, but the four-curve table also shows it can whipsaw when the arm threshold is set below the curve's pre-run equity, so the mechanism has a configuration risk of its own that a live deployment would have to manage.

The carry sleeve rests on a longer six-year sample and is the more general result, but even there the read that survives scrutiny is a net Sharpe near 1.0 with a 46 percent drawdown, three losing years, and a bootstrap interval whose lower bound is barely above zero. Its steady component, the funding coupon, is only six to fifteen percent a year, and the rest is a crowded-name price spread that carries crowding risk and drove every one of its losing years. Anyone who would deploy this runs into limits the simulation does not show. The account figures carry a flat 0.1 percent round-trip fee on the directional side and an explicit tiered slippage model on the carry side, but neither models market impact at size nor the capacity limit where the carry spread is arbitrated away by others running the same screen, and both are real constraints before any meaningful capital is committed. The ratchet and loss filter are studied on simulated account curves, not live fills; a live trailing stop faces slippage on the stop itself, gaps through the stop level, and the operational question of whether it acts on marked or realized equity, none of which the daily-close simulation captures. The reconciliation of a live risk-managed book against its backtest, in the spirit of the divergence the program saw elsewhere, is left to follow-on work. The most useful extension is not a fourth mechanism but a longer live record of these three, across a regime that includes a sustained uptrend, to see whether the loss filter and the ratchet earn their keep when the beta they lean on runs the other way.

11. Conclusion

The usual response to a forecast that fails is to reach for a better forecast. The companion papers argued that for these instruments the better forecast is not available, and this paper takes that conclusion seriously enough to ask what is left. The answer is risk control, and it is worth having on its own terms, as long as each piece is read against its own boundary. A loss filter that declines the worst entries turned a plus 13.2 percent month into plus 87.8 percent by subtracting

trades, and a permutation test and a monotone entry-versus-outcome relationship show the selection is real rather than lucky, while the model's own confidence is shown to be no filter at all. A profit-armed ratchet kept most of a windfall it could not predict, locking about 2,926 dollars where an unmanaged book peaked at 3,461 and then gave back all but 1,878, identically across a broad plateau of settings, though it whipsaws when armed too low. A market-neutral carry sleeve earned a net Sharpe near 1.0 over six years, chosen over a momentum factor that dies out of sample, and reported with its three losing years, its wide bootstrap interval, and its small steady coupon, while its favorable-window mirage is set aside. Not one of the three predicts anything, and none of them holds up in every regime. Together they amount to a bounded, deployable structure rather than an edge that prints. This is the part of the program that holds up over time, and it does so precisely because it never relied on knowing which way the next move would break.

Author Contributions and Use of AI Tools

The research questions, experimental design, hypotheses, methodological choices, and interpretation of results in this paper are the author's own. All software implementation, including the account simulator, the loss filter and trailing-stop logic, the cross-sectional carry backtester, the permutation and block-bootstrap tests, the cost and slippage models, the reproduction scripts, and the figures, was written in Python with the assistance of Claude, a large language model by Anthropic, acting as a coding assistant under the author's direction. Claude is a software tool and is not an author of this work. The author is solely responsible for the content, the claims, and any errors.

Data and Code Availability

Every quantitative claim in this paper is reproduced by two deterministic scripts that read the program's data and print each number quoted above. The first computes the full six-year carry stress test with its per-year breakdown and breakeven-slippage sweep, and the profit-armed ratchet grid over the directional account curves. The second computes the loss-filter permutation test and the entry-extreme monotonicity from the resolved-trade record, the ratchet across all four account curves and the wide parameter grid, the expected-best-of-grid deflation and stationary block-bootstrap interval for the carry Sharpe, and the funding-versus-price-spread

decomposition. The favorable-window carry configuration is reproduced by a third script. The directional account series and resolved-trade record, the carry price and funding data, the scripts, and the figure code are available from the author on reasonable request, subject to the terms of the third-party data sources used to build the features, with a companion code repository in preparation. The directional book covers twenty Binance USD-margined perpetual contracts over June 2026; the carry backtest uses fifteen-minute prices and realized funding for the same universe from 2021 through mid-2026.

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